Least cost perfect pairing method And the proof of p=np

Abstract

in the subject of math and computer science, traveling salesman problem is known to be a very difficult problem. whose solution is easy to find. But as the size of the problem increases, it becomes very difficult to solve it in a short time.
But through this paper, I have presented the best way to solve TSP in Polynomial time, by the help of the " LEAST COST PERFECT PAIRING" method, Which will prove P=NP by solving a 50 year old problem P Vs NP.

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Keyword.

Tsp,polynomial time,perfect pair,starting city pair (SCP), least cost middle city pair LC (MCP),
Least cost Returning city pair LC (RCP)

Introduction.

The Traveling Salesman Problem belongs to the class of NP-Hard problems in combinatorial optimization.[1]

The problem is:

"Given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city exactly once and returns to the origin city?" [1]

Here we know that Hamiltonian Tour exists because the Hamiltonian cycle problem is to find if there is a tour that visits every city exactly once. [3]

There are many well known method for solving TSP,ex-brute-force method,dynamic programming method,branch and bound, ,greedy method,nearest neighbors method, etc. [1]

Where greedy methods and nearest neighbor methods are computationally inefficient, they may not

lead to optimal solutions.

Whereas in the brute-force approach, we need to evaluate every possible tour and select the best one. For n number of vertices in a graph, there are (n - 1) number of possibilities.

Hence, it is not useful in a large number of cities.

Instead of brute-force when we use the dynamic programming approach, the solution can be obtained in less time, having a time complexity of O(n22n), where n = the number of vertex/nodes of the graph. though not give a polynomial time solution! The TSP is very important in theoretical computer science and operational research. If we can find a polynomial algorithm for TSP, then P = NP. Here, P denotes the class of decision problems solvable by some algorithm in polynomial time. The notation NP stands for "nondeterministic polynomial time," and the class of questions for which an answer can be verified in polynomial time is called NP [4].

will be proved, then all of the NP problems will be solved deterministically in Polynomial time. With this, many areas like finding better drugs, cancer treatment,

accurate prediction of natural phenomena, many engineering problems, vehicle routing, circuit design, graph optimization,cryptography, etc. will become easier. But despite all the efforts of the researchers,
The solution of very large TSP was not found in polynomial time,but this paper, I have
presented the polynomial solution of TSP by "least cost perfect pairing" method, which also proves P=NP

Background

Under the TSP problem, there are a set of N cities in front of the salesman, now the
salesman has to travel to a number of cities, while the cost between each city is given. the salesman has to choose the least cost route , but the problem arises when there are more cities in front of him.
For ex-There are 100 cities in front of salesmen, and salesmen start their journey from city 1. After traveling to the rest of the cities and coming back to City 1, choose the city with the lowest cost among the salesmen (N-1). permutation is required,

where n=100, then 100-1!=99!=9.33262154e155 route will be generated. Which is impractical!
and it is impossible to find the optimum root in polynomial time as the size of N numbers of cities increases by the methods known so far.
But the salesman can find the optimum route for millions and trillions of cities in polynomial time using the "least cost, perfect pairing" method I developed!

Methodology

First step.
Under TSP, N×N cost adjacency matrix is given for N number of cities.
So first, we divide the matrix into 3 parts
1-starting city pair part (SCP)
2- middle city pair part (MCP)
3- returning city pair Parr (RCP)

Look at the given example matrix 1

Example matrix-1

	A	B	C	D	E
A	∞	2	5	7	1
B	6	∞	3	8	2
C	8	7	∞	4	7
D	12	4	6	∞	5
E	1	3	2	8	∞

1 has

1- Starting city pair part=AB, AC,AD,AE

2-Middle city pair part=BC,BD,BE

CB,CD,CE

DB,DC,DE

EB,EC,ED

3-returning city pair part=BA,CA, DA, and EA

Second step.

Now the reduced matrix of starting city pair part (SCP), least cost middle city pair LC (MCP) and least cost returning city pair LC (RCP) have to be made.

from the given matrix

Reduce matrix -1

	A	B	C	D	E
A	∞	2	5	7	1
B	6	∞			2
C			∞	4	
D		4		∞	
E	1		2		∞

It has

1-starting city pair part (SCP) =AB,AC,AD,AE

2-least cost middle city pair part

LC(MCP)=BE,CD,DB,EC

3-Least cost returning city pair part LC (RCP) = EA,BA

NOTE:-Least cost Returning pair city LC (RCP) is to be taken once considering AB starting pair city, second time keeping AC ... AN in mind. If the list of LCs (RCPs) is not same, then at least two
LC (RCP) have to be selected.
In given example of matrix 1,
EA, BA are two LC (RCP)

2.1-After reducing the matrix once, we also have to reduce LC (MCP)

For ex-SCP=AB
LC(MCP)=BE,CD

then:-AB to BE=2
AB to CD = 3

BE to CD = 2
CD to E = 5
Now we have two least cost perfect pair is
AB+BE=2
BE+CD=3

So the root from SCP, AB, becomes like this.

AB+BE+CD+A=ABECDA

For SCP,AC

LC(MCP)=BE,CD,DB

So

AC to CD = 4

CD to DB = 4

DB to BE = 2

So the root from SCP, AC becomes like this.

AC+DB+BE+A=ACDBEA

NOTE- If the cost of CD to BE is to be found, then CD is to be taken as D, and BE is to be B.

Ex-

CD to BE

D to B = 4

but if If the cost of BE to CD is to be found, then BE is to be taken E and CD is to be C.

BE to CD

E to C = 4

And if the least cost perfect pair is like this

BC+CD+DE+EG+GH=BCDEGH

(BCDEGH known as a group of LCMCP.)

And another LC (MCP) is KL

Then the cost of

BCDEGH to Kl mean,

H to K.

but if If the cost of KL to BCDEGH is to be found,

Then

KL to BCDEGH mean,

L to B.

But if SCP is AB, and we need to find cost from KL to BCDEGH,

then consider KL as L and BCDEGH as C,not "B"

AB+KL+CDEGH=ABKLCDEGH

Because B city is repeating here.

Third step.

Now SCP, LC (MCP), and LC (RCP) are to be combined to make a Route N number for the city.

The following points have to be kept in mind while preparing the route.

3.1 The least cost perfect pair method works on the principle that if a salesman starts his journey from AB, ACAD, or AN, he will visit a group of least cost middle city pairs or single middle city pairs and not every city, as explained in the brute force method and dynamic approach, in which even the high cost city has to be visited.

For example-

SCP=AB

LC(MCP)=BC,CD,DE,FG

LC(RCP)=GA

The group of LC (MCP) for SCP, AB, will be like this-

BC+CD+DE =BCDE -1 groups

FG -2nd group

then the route will be likis:-

AB+BCDE+FG+A=ABCDEFGA -1st route.

AB+FG+CDE+A =ABFGCDEA -2nd route.

3.2-In the above example, LC (MCP) is divided into two groups, so it is easy to permute, ,but if the LC (MCP) is in more than two groups, it will be solved in this way:-

For example, Salesman has to visit 8 cities,

A,B,C,D,E,F,G,H

So the route for SCP,AB will be as follows-

Suppose,

LC(MCP)=BH,CE,DG,HF,FC,GB,

LC(RCP)=CA

Then

BH+HF+FC+CE will be the first group

DG will be the second LC (MCP), and I will be the third single city.

Now we have three groups of LC (MCP), HFCE,DG, and I.
Now, before making the optimum route for SCP AB, the minimum cost between these groups has to be found from the matrix.

For example, if the cost of city pairs in the matrix is as follows,
FOR HFCE
HFCE to DG = 5
HFCE to I=8

FOR I
I to HFCE = 3
I to DG = 7

FOR DG

DG to I=4

DG to HFCE = 8

we know we have three least cost perfect pair

1-HFCE+DG=5

2-I+HFCE=3

3-DG+I=3

So their route will be like this.

1.ABHFCE+DG+I+HE+A

2.AB+HFCE+DG+I+A

Note

Once a group of city pairs is formed, we will measure the cost from the first and last city of a larger group while taking the cost for the next city pair or single city pair from that group.

In the above example, while taking the cost of BHFCE to DG, we will measure the cost of E to G.

Whereas if the cost of DG to BHFCE is to be measured, then the cost of G to B will be measured.

3.3- If the group of LC (MCP) is formed as
BC+CD+DE
So it means B to C, C to D and D to E
at least distance or cost.

2-If
SCP=AB,AC,AD,AN
LC(MCP)=BC,CD,DE,

Then
AB+BC=ABC
AD+DE=ADE
AC+CD=ACD

and if the group of LCs (MCPs) is such that
AB+BC+CD+DE+FG+A

so the route will be like this
ABCDEFGA.

And if the order of pairs of this group is changed, or "AB+FG is done, then

The root of BC+CD+DE will be like this.

AB+FG+CDE+A
ABFGCDEA

3.4-.If LC(MCP)=BC,CF,DB,EK
LC(RCP)=BA,CA

So if we will make route with
AB+LC(MCP)+LC(RCP)

So we will not include "DB" while making root from AB because it will recycle cities.
AB+BC+CF+DB+BA
Here, city B is repeating.

This is what to keep in mind while making routes from SPC,AC, AD, and AN

3.4-The route of n number of cities is to be made once, keeping in mind the starting city pair(SCP)+low cost middle city pair (LC) (MCP).

ex-,SCP=AD
LC(MCP)=BE,CD,DB,EC
LC(RCP)=EA

Then
AD+DB+BE+EC+CA=ADBECA

3.5-For the second time, starting city pair (SCP) +low cost middle city pair (LC) + low cost
Returning city pair LC (RCP) has to be made the route of a number of city.
Ex-
AD. EA
AD+DB +C+EA

AD+C+DB+EA

NOTE

from above to take care of this

Here, LC (RCP) = EA

So while making a route to SCP and LC (RCP), we will not make a route from EC to the same city.

because this is repeating the city E.

AD+DB+EC+EA

3.6-If all LC (MCP) and LC (RCP) are included while making route from SCP, then will stop making route from that SCP further.

EX.

SCP=AB

LC(MCP)=BC,CD,DE

RCP=EA

Then

AB+BC+CD+DE+EA

ABCDEA

3.4-Do not leave any LC (MCP) or LC (RCP) while creating the route!

For Ex-

SCP=AB

LC(MCP)=BC,CD,EF,DC

then-

AB+BC+CD+EF+A. 1 ROUTE

AB+EF+BC+CD+A. 2 ROUTE

Now that we have LC (MCP) , DC left, we have to make routes from there too.

AB+DC+EA 3 ROUTE

Fourth step.

reduce the LC (MCP)

lC (MCP) has to be reduced!!

Which means if the salesman now starts his journey from AB,AC, AD, and AN, he will always try to visit such LC (MCP) between which the cost is minimum.

And which LC (MCP) has the least cost? depending on whether they are in the form of BC+CD+DE or XN+NY+YM+MZ.

We understand this from the given example matrix-
In given example matrix, example 1,
We found:-
starting city pair (SCP) = AB,AC,AD,AE
Low cost middle city pair LC (MCP) = BE,CD, DB, EC
Low cost returning city pair LC (RCP) = EA,BA
Now we will make a route by combining a starting city pair, a low cost middle city pair,and a low cost returning city pair.

SCP+LC(MCP)+LC(RCP)=AB,AC,AD,AE,+BE,CD,DB,EC+EA,BA
SCP+LC(MCP)+LC(RCP) Perfect combination will give the following route
For starting city pair AB:-
SCP=AB
LC(MCP)=BE,CD,DB,EC

we know we have the low cost middle city pair

LC(MCP)= BE,CD,DB,EC

and Low cost Returning city pairs

LC(RCP) =EA,BA

But to avoid city repetition, BA and DB city pairs will not be included in the route construction.

Then we have the following city pairs for SCP AB will be left

LC(MCP)=BE,CD,EC

LC(RCP)=EA

the following least cost perfect pairs will be formed for SCP,AB

SCP=AB+LC(MCP)=BE+CD+EC+LC(RCP)=EA

1- AB+BE+EC+DA=ABECDA

2- AB +EA

AB+CD+EA=ABCDEA

For starting city pair, AC:-

SCP=AC

LC(MCP)=BE,CD,DB,EC

we know we have the low cost middle city pair

LC(MCP)= BE,CD,DB,EC

and Low cost Returning city pairs

LC(RCP) =EA,BA

But to avoid city repetition, EC city pair will not be included in the route construction.

Then we have the following city pairs for SCP AC will be left

LC(MCP)=BE,CD,DB

LC(RCP)=EA

the following least cost perfect pairs will be formed for SCP,AC

SCP=AC+LC(MCP)=BE+CD+DB+LC(RCP)=EA

3- AC+CD+DB+EA=ACDBEA

For starting city pair, AD:-

SCP=AD

LC(MCP)=BE,CD,DB,EC

we know we have the low cost middle city pair LC (MCP) = BE,CD, DB, and EC

And Low cost returning city pairs

LC (RCP) =EA,BA.

But to avoid city repetition, CD city pair will not be included in the route construction.

Then we have the following city pairs for SCP AD will be left

LC(MCP)=BE,EC,DB

LC(RCP)=EA

4- AD+DB+BE+EC+CA=ADBECA

5- AD+DB +C+EA

AD+DB+C+EA=ADBCEA.

6-AD+C+BE+EA=ADCBEA.

For starting city pair, AE:-

SCP=AE

LC(MCP)=BE,CD,DB,EC

we know we have the low cost middle city pair LC (MCP) = BE,CD, DB, and EC

and Low cost Returning city pairs

LC(RCP) =EA,BA

But to avoid city repetition,BE and EA city pairs will not be included in the route construction.

Then we have the following city pairs for SCP AE will be left

LC(MCP)=CD,EC,DB

LC(RCP)=BA

7- AE+EC+CD+BA=AECDBA

Fifth step.

Now the cost of each route has to be determined, and the route whose cost is less will be the optimum route for the salesman.

Example From Matrix1 solution, we get-

AB+BC+CD+DE+EA=ABCDEA-1

2+3+4+5+1=15

AB+BE+CD+DA=ABECDA-2

2+2+3+4+12=23

AC+DB+EA=ACDBEA-3

5+4+4+2+1=16

AD+DB+BE+CA=ADBECA-4

7+4+2+2+8=23

AD+C+BE+EA=ADCBEA-5

7+6+7+2+1=23

AD+DB+C+EA=ADBCEA-6

7+4+3+7+1=22

AE+EC+CD+DB+BA=AECDBA-7

1+2+4+12+6=25

So for salesman, route 1, ABCDEA = 15.

Is optimum route

Thus, we see, 7 routes are required for 5 city to find the optimum route by "perfect pairing method"

At the same time, there was a need for 5-1! =24 permutations of the city ,by" Brute force method."

see example matrix 2

	A	B	C	D	E	F
A	∞	20	23	27	29	34
B	21	∞	19	26	31	24
C	26	28	∞	15	36	26
D	25	16	25	∞	23	18
E	23	40	23	31	∞	10
F	27	18	12	35	16	∞

Reduce matrix 2

	A	B	C	D	E	F
A	∞	20	23	27	29	34
B	21	∞	19			
C			∞	15		
D		16		∞		
E	23				∞	10
F			12			∞

Now we have

SPC=AB,AC,AD,AE,AF

LC(MCP)=BC,DB,EF,FC,CD

LC(RCP)=BA,EA

FOR SCP=AB

AB+BC+CD+EF+FA=ABCDEFA. -1

20+19+15+23+10+27=114

AB +EA

AB+FC+CD+E+EA =ABFCDEA -2

20+24+12+15+23+23=117

FOR SCP=AC

AC+CD+DB+EF+FA=ACDBEFA 3

23+15+16+31+10+27=122

AC+CD+EF+BA=ACDEFBA. -4

23+15+23+10+18+21=110

AC. +DB+BA

AC+EF+DB+BA=ACEFDBA -5

23+36+10+35+16+21=141

FOR SCP=AD

AD+DB+BC+EF+FA=ADBCEFA. -6

27+16+19+36+10+27=135

AD. +BA

AD+EF+FC+BA=ADEFCBA -7

27+23+10+12+28+21=121

AD+DB+EF+FC+CA=ADBEFCA. -8

27+16+31+10+12+26=118

FOR SCP=AE

AE +BA

AE+EF+FC+CD+DB+BA=AEFCDBA -9

29+10+12+15+16+21=103

AE+EF+BC+CD+DA=AEFBCDA -10

29+10+18+19+15+25=116

FOR SCP=AF

AF+FC+CD+DB+EA=AFCDBEA. -11

34+12+15+16+31+23=131

AF+FC+CD +BA

AF+FC+CD+E+BA=AFCDEBA -12

34+12+15+23+40+21=145

The optimum route for salesmen is- AEFCDBA=103.

NOTE-Once again, we know in detail the perfect pair of route no. 11 and 12

we know SCP=AF

LC(MCP)=BC,DB,FC,CD,EF (To avoid repetition of the city, EF will not be included in pairing.)

LC(MCP)=BA.

When we pair with EF

In the first instance, all LCs (MCPs) are paired except EF-Look route No. 11

the 12th root will be formed from the remaining pair BC.

Thus, we see, 12 routes are required for the 6 cities to find the optimum route by "perfect pairing method"

At the same time, there was a need for

6-1! =120 permutations of city, by" Brute force method."

In brief, the salesman can choose the optimum route using the three steps mentioned below.

1st step.

1A. SCP, LC (MCP), and any two LCs (RCP) have to be taken out of the matrix.

1B. If the last city of the LC (MCP) is the same, then the second LC (MCP) is to be included.

If they are not identical, then only one pair of LC (MCP) is to be taken out.

1C. Similar is to be done with RCP.

2nd step.

2A. Route to N cities is to be made with SCP and LC (MCP) in such a way that the cost between them is minimum; by doing this, the salesman will return to his home city A.

2B. At the same cost between pairs, more routes will be built, and the salesman will return to his home city.

2C. If LC (RCP) is also included while making the route of SCP and LC (MCP), then there is no need to make a separate route for LC (RCP).

3rd step.

3A. Now, with SCP, LC (MCP) and LC (RCP) routes have to be made for N cities.

3B. If pairs with equal cost are formed, more routes will be formed, otherwise, only one route will be formed.

Time complexity of the "least cost, perfect pairing" method.

From the given matrix, example 1

n = 5

K+k+k+k+k+k+K+K= 7 routes are required

When n = 6,

K+K+K+K+K+K+K+K+K+K=12 routes are required.

Thus, if city=N,

then to get the optimum route, the route will always be less than 5N (in best case)

in worst case scenario

If city=N

then to get the optimum route, the route will always be less than =10N or it will depend on

LC (MCP) and LC (RCP) have the same cost.

but in any case, complexity will be below

O(n2)

Thus, the time complexity will be always O(N)

In "least cost Perfect pair method, we will see the worst case with matrix example 3

Example matrix problem

reduce matrix example 3

	A	B	C	D	E	F
A	0	20	23	27	29	34
B	21	0	19			24
C	20		0	20		
D	21	22	19	0		
E	21		19		0	26
F		29	19			0

Here

SCP=AB,AC,AD,AE,AF

LC(RCP)=BA,DA ,EA,CA

1-LC(MCP) =BC,DC,EC,FC

2-LC(MCP)=BF,DB,EF,FB

NOTE:- Here, LC (MCP) and LC (RCP) are more important because everyone's COST here is almost the same!

Therefore, to extract the optimum root, we have to include the second LC (MCP) in the matrix to form the route.

For SCP=AB

The route will be-

AB+BC+CD+DE+FA=ABCDEFA -1

AB+BF+FC+CD+EA=ABFCDEA -2

AB+BF. +EC+CA

ABFDECA. -3

AB+EF+DC+CA=ABEFDCA -4

(Due to more LC (MCP), more routes than AB)

Important - If the LC (MCP) and LC (RCP) of the same cost are higher, then more and more routes will be formed to get the optimum route.

This would be the worst case under the " LEAST COST PERFECT PAIRING" method.

With this, we will see that as we create routes for n cities by increasing the cost of the least cost middle city pair LC (MCP), the cost of the route will increase.

This is proving that the lower the cost of LC (MCP), the lower the cost of the route.

Proof-In this way, solving the above example matrix 1, 2, and 3, we know that we can find the
optimum root in Polynomial time only by the pair of SCP, LC (MCP) and LC (RCP)
And also know, if we have given matrix,
A N-1 city pair will be required to extract the optimum route. (in best case)
and in the worst case (N×2-2) city pairs are required to extract the optimum route!

Conclusion.

Known exact and metaheuristic algorithm fail to solve optimal solutions of Tsp in Polynomial
time.
That's why I have created a new method called "LEAST COST PERFECT PAIRING" to deal with this problem, Under which, by combination of SCP+LC (MCP)+LC (RCP), we can get the optimum solution in

polynomial time by taking the least root, which the known methods so far have failed to do. The
The proposed method also proves P=NP by solving TSP in polynomial time O(N) or, in the worst case, in less than O(n2).
Which will pave the way for better solutions of math,computer science, medical science,,
economy, and other unsolved NP problems by reducing the complexity of computers.

Bibliography.

1. Sangwan,Shabnam,"Literature Review on Traveling Salesman Problem," International
Journal of Research, June 2018
2. Fischer, Thomas, and Merz, Peter,"Reducing the Size of Traveling Salesman Problem
Instances by Fixing Edges, September 2008
3.Datta, Subham, "Traveling Salesman Problem – Dynamic Programming Approach,"
baeldung.com, 11 November 2022
4.Wang,ziliang,"Fast Algorithm for the Travelling Salesman Problem and the Proof of P =

NP,"Research institute for ESMD Method and its Applications,college of science, Qingdao University of Technology,Qingdao,China.12 december 2018